

The ND280 semi-frequentist fitting strategy

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In a semi-frequentist approach, a prior covariance matrix for each parameter set (flux, detector and cross-section parameters) is provided. The ND fitting strategy implies finding the maximum likelihood () or the minimum of $\chi^2 = -2\ln$ by using a gradient descent method in such a way to ensure that ND event rate predictions match data. One of the requirements for function is to be smooth and continuous. ND fit utilizes a gradient descent method implemented in Minuit2 [?] which is a part of the ROOT analysis framework [?].

The uncertainties are determined by inverting a Hessian matrix, which includes the second derivatives of the likelihood function, using the Minuit2 HESSE algorithm. The best-fit values of the parameters related to far detector, along with their covariance matrix, are then carried forward to the next step of the sequential fit process, far detector fit. Where far detector fit is performed using VaLOR or P-Theta semi-frequentist frameworks.

1 The ND280 likelihood

The likelihood for ND280 can be described by Poissonian distribution since the ND280 data is of the counting type separated into bins. Poissonian gives a probability to observe N_i^{obs} events in the i th bin for predicted N_i^{pred} events in the same bin.

$$= \prod_{i=1}^N \frac{(N_i^{pred})^{N_i^{obs}} e^{-N_i^{pred}}}{N_i^{obs}!} \quad (1)$$

where N_i^{pred} depends on some set of parameters $\vec{\theta}$.

In the statistical analysis, it is usual to use ratio in order to normalize the likelihood function to nominal predictions. It allows the cancellation of the factorial term.

$$ratio = \frac{\prod_{i=1}^N (N_i^{pred})^{N_i^{obs}} e^{-N_i^{pred}}}{\prod_{i=1}^N (N_i^{obs})^{N_i^{obs}} e^{-N_i^{obs}}} \quad (2)$$

Furthermore, it's easier to calculate a likelihood by moving from product to sum by using χ^2 variable that is equal to $-2\ln$. Thus in terms of $\Delta\chi^2 = \chi^2 - \chi_{nom}^2$ that is referred to *ratio* the Poissonian distribution is taking the following form:

$$\Delta\chi^2 = 2 \sum_i (N_i^{pred} - N_i^{obs} + N_i^{obs} \ln \frac{N_i^{obs}}{N_i^{pred}}) \quad (3)$$

N_i^{pred} refer to MC simulated events while N_i^{obs} can be considered as reconstructed data in ND280. The use of a semi-frequentist approach assumes an accounting for parameters' prior uncertainties. This prior information is included in order to ensure that parameter values will not deviate a lot from their nominal values. Therefore, an additional term is added to a Poissonian or statistical, namely the systematics or penalty term. The $\Delta\chi^2$ can be expressed as follows:

$$\Delta\chi_{total}^2 = \Delta\chi_{stat}^2 + \Delta\chi_{sys}^2 = 2 \sum_i (N_i^{pred} - N_i^{obs} + N_i^{obs} \ln \frac{N_i^{obs}}{N_i^{pred}}) + \sum_p^{N_{par}} \sum_q^{N_{par}} \Delta\vec{\theta}_p (V_\theta^{-1})_{p,q} \Delta\vec{\theta}_q \quad (4)$$

where the penalty term expression is based on the assumption that priors have a Gaussian shape. $\vec{\theta}$ is a set of parameters and V_{θ} is a prior covariance matrix. $\Delta\vec{\theta}$ reflects a parameter difference with respect to the nominal value.

In ND280, three uncorrelated models are used as input: flux, detector and cross-section. Each of the models is introduced by a set of parameters. Penalty term for ND280 in this case can be introduced by the sum of such terms for each model, where $\vec{f}$, $\vec{d}$, $\vec{x}$ refer to flux, detector and cross-section parameters, respectively.

$$\begin{aligned}\Delta\chi_{sys}^2 = & \sum_p^{N_{par}} \sum_q^{N_{par}} \Delta\vec{f}_p (V_f^{-1})_{p,q} \Delta\vec{f}_q + \\ & + \sum_p^{N_{par}} \sum_q^{N_{par}} \Delta\vec{x}_p (V_x^{-1})_{p,q} \Delta\vec{x}_q + \\ & + \sum_p^{N_{par}} \sum_q^{N_{par}} \Delta\vec{d}_p (V_d^{-1})_{p,q} \Delta\vec{d}_q\end{aligned}\quad (5)$$

Starting from OA 2020 the Barlow-Beeston method [?] was implemented in order to account for the MC statistical uncertainty coming from the difference between generated MC and true MC (when the infinite amount of events are simulated). Consequently, the statistical part of likelihood should have the following form:

$$\Delta\chi_{stat}^2 = \Delta\chi_{Poisson}^2 - \Delta\chi_{MC\ stat}^2 \quad (6)$$

where $\Delta\chi_{MC\ stat}^2$ is a MC statistical uncertainty or MC penalty term.

In order to account for the MC penalty term, each bin should be scaled by the parameter β_i in order to obtain the correct number of events $N_i^{true} = \beta_i \times N_i^{pred}$. However, such implementation is cumbersome and requires the introduction of a big amount of additional parameters associated with each bin that will be varied in the fit. Instead, a Gaussian distribution is assumed for β_i and minimizing $-2\ln$ with respect to β_i , the β_i can be found analytically for each bin by solving an expression:

$$\beta_i^2 + (\mu\sigma_{\beta_i}^2 - 1)\beta_i - n\sigma_{\beta_i}^2 = 0 \quad (7)$$

where μ is a number of simulated events N_i^{pred} in the certain bin with all the weights w_i applied to the event in this bin; σ_{β_i} is associated to the bin statistical uncertainty and can be expressed as $\sigma_{\beta_i} = \sqrt{\sum w_i^2 / N_{MC}^{gen}}$, where w_i is the sum of the weights in bin i and n is the number of observed data events in this bin N_i^{obs} .

Consequently, in this form, the Barlow-Beeston MC penalty term can be simply added to the Poissonian term and the statistical term has the following form:

$$\Delta\chi_{stat}^2 = 2 \sum_i^N [(N_i^{pred} - N_i^{obs} + N_i^{obs} \ln \frac{N_i^{obs}}{N_i^{pred}}) + \frac{(\beta_i - 1)^2}{2\sigma_{\beta_i}^2}] \quad (8)$$

The minimum $\Delta\chi^2$ in general form can be expressed as:

$$\Delta\chi_{min}^2 = \sum_p^{N_{par}} \sum_q^{N_{par}} (\theta_p - \theta_p^{min})(\theta_q - \theta_q^{min}) \frac{\partial^2 \Delta\chi^2(\theta^{min})}{\partial\theta_p \partial\theta_q} \quad (9)$$

where $\frac{\partial^2 \Delta\chi^2(\theta^{min})}{\partial\theta_p \partial\theta_q}$ is a Hessian matrix.